

HOW PLASMA THERAPY WORKS IN COVID-19 PATIENTS?

When a pathogen such as SARS-CoV-2 infects the human body, the immune system kicks into gear and begins producing antibodies to combat the invaders (the virus). If a person has recovered from this virus, the antibody is found in the plasma, the liquid that contains blood cells. In the plasma therapy, the plasma containing the antibody is harvested from the blood of the recovered patient. The plasma is then separated from the blood and injected into an infected person's body. This antibody creates passive immunisation (immediate, short-term immunity) and helps them fight the virus.

ment of India and Member of Niti Aayog. According to Lav Agarwal, Joint Secretary, Ministry of Health and Family welfare, members of the task force include Ministry of AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha and Homoeopathy), ICMR (Indian Council of Medical Research), DST (Department of Science and Technology), DBT (Department of Biotechnology), CSIR (Council of Scientific and Industrial Research), DRDO (Defence Research and Development Organisation), DG-Health Services and Drug Controller General of India. The DBT will act as the central coordinating authority of the task force with its primary goal being setting up a process for vaccine development. The task force will closely monitor the national and international organisations that are working on a vaccine for COVID-19 and will also work on collecting bio-specimens to help with future trials of drugs and vaccines.

EXPERIMENTAL TREATMENTS AND CLINICAL TRIALS

While research and trials are in progress to develop a vaccine for COVID-19, there is a dire need for treatment since the novel coronavirus has killed over 19 lakh people globally. As researchers continue to conduct extensive experiments to come up with a potential treatment for COVID-19, doctors and scientists have resorted to repurposing existing drugs or a making a concoction of existing drugs typically used to treat other viral diseases.

Plasma therapy

One of the most promising treatment options for the viral infection that is being explored currently is 'convales-

cent-plasma therapy'. The FDA has officially given the nod to Columbia University to launch clinical trials to determine whether blood plasma taken from COVID-19 survivors can be utilised to prevent infections in high-risk individuals. India joined the list of countries using plasma therapy early in April. Kerala, Karnataka, Delhi, Gujarat and Maharashtra, among other states, have begun clinical trials for plasma-based treatments.

The ICMR in India recently permitted states to carry out clinical trials of this experimental procedure only to treat severely-affected COVID-19 patients. The treatment is not yet recommended as a standard one and clinical trials are being carried out to corroborate the effectiveness. On April 24, the Delhi government announced that it is aggressively pushing for plasma therapy and that it is beginning to show "encouraging results". If the clinical trials do perform favourably, the Centre's approval will be sought, for plasma therapy to be used on all serious COVID-19 patients. This move comes after the



Plasma therapy could be an effective method to treat critically-ill COVID-19 patients

improvement in the health conditions of a 49-year-old male who was given plasma therapy at Max Hospital, Saket. The patient was successfully taken off ventilator support after the administration of plasma therapy, according to Max Hospital. Plasma therapy was actually used to treat patients back in the 2003 SARS epidemic, the 2009 H1N1 outbreak and in 2014 during the Ebola outbreak. The ICMR also gave Kerala the nod to conduct clinical trials of plasma treatment. However, the state cannot implement this just yet since it has no critically-ill patients as of writing this article. The state has been lauded by healthcare professionals, the government and the citizens for managing the epidemic spread efficiently.

Remdesivir

The ICMR has recommended numerous potential drugs to fight COVID-19 in India. Among them is the antiviral drug Remdesivir, an experimental drug developed by Gilead Sciences Inc., that was used during the Ebola outbreak but failed. The inclination to use this drug to treat COVID-19 is due to a study published in the New England Journal of Medicine which stated that two out of three critical COVID-19 patients (dependant on oxygen support and/or ventilators) that were administered Remdesivir showed definite signs of improvement. Indian drugmakers such as Cipla, Glenmark and Dr Reddys are already preparing to develop Remdesivir after, hopefully, licensing it from Gilead Sciences.

However, the companies can still formulate the drug before being granted licensing provisions as per the 'Bolar Provision', strictly for research and development purposes. Pharma companies have begun developing the drug's raw material, the APIs (active pharmaceutical ingredients). A Glenmark spokesperson told The Print that once Gilead approves the licensing, Indian companies could start supplying the drug within a month. Unfortunately, the promising antiviral drug Remdesivir allegedly failed its first clinical trial, according to surfacing reports. A randomised